
EDUCATION

Calgary, AB **University of Calgary** **Fall 2019 – December 2025**

- **Double Major:** Software Engineering, B.S.E; Business, M.S (*Overall GPA: 3.7*)
- **Undergraduate Coursework:** Operating Systems; Algorithms and Data Structures; Networks; Comp. Architecture; Machine Learning; Circuits; Logic Design; Signal Processing; Compilers; Database Design, Graphics.

EMPLOYMENT

Software Engineer, Founder **NextRipple, Calgary** **09/2023 – 09/2025**

- **Founded NextRipple**, leading both **technical and business strategy** to develop consumer tech solutions.
- **Designed and built** multiple consumer-facing applications, including:
 - A **product finder app** (similar to Uber Eats) enabling a **unified multi-retailer checkout experience**.
 - A **fitness app** integrating **ML-powered workouts, AI-driven diet plans, and computer vision body analysis**, along with **data management and rewards features**.
- **Led end-to-end product development**, from **UI/UX and engineering** to **market validation and MVP deployment**.
- **Managed business operations**, securing **\$250,000+ in funding**, handling **financial modeling, investor pitches, and strategic growth**, scaling to **20 employees**.
- **Built key partnerships** with **investors, stakeholders, and clients**, aligning innovation with market needs.

Software Engineer, Intern **Appara, Vancouver** **01/2023 – 11/2023**

- Designed and developed a **legal document automation platform**, reducing manual effort by dynamically generating **corporate and real estate legal documents** from minimal inputs.
- Led **UI/UX design**, creating **cohesive themes, interactive prototypes, and accessible interfaces**, improving user experience and adoption.
- Built **scalable front-end components** using **HTML, CSS, JavaScript, and C#**, following **OOP principles, MVC architecture, and modular design** for maintainability.
- Implemented **secure authentication, optimized database queries, and integrated backend APIs**, enhancing **performance, security, and compliance with legal data regulations**.
- Collaborated with **lawyers, developers, and stakeholders**, aligning technical solutions with business goals while following **Agile methodologies, CI/CD pipelines, and best software practices**.

Automation Engineer, Intern **Engineered Air, Calgary** **01/2022 – 01/2023**

- Engineered **HVAC computation software**, leveraging **C#, C, C++, JavaScript, React, and Node.js** to automate complex **thermodynamic and mechanical calculations**.
- Applied **OOP, modular architecture, and real-time data processing**, optimizing **engineering workflows** and enhancing **system modeling precision**.
- Integrated **algorithmic optimizations** to streamline **performance simulations**, ensuring **scalability and computational efficiency**.
- Bridged **software and mechanical systems**, collaborating with engineers and conducting **on-site factory analysis** to align **digital solutions with real-world HVAC applications**.

ML Engineer, Intern **Stratuscent, Quebec** **08/2021 – 04/2022**

- Engineered an **AI-driven smoke detection system** for commercial kitchens, integrating **embedded systems, computer vision, and edge computing** for real-time safety.
- Developed **real-time image processing** using **Python, TensorFlow, OpenCV, and PyTorch**, deploying **CNNs** on edge devices for **low-latency detection**.
- Designed and optimized **firmware in C, C++, Embedded Python**, utilizing **sensor fusion** for data from thermal, gas, and optical sensors to ensure accuracy.
- Architected **IoT connectivity** with **MQTT, WebSockets**, and a **Node.js/React dashboard**, enabling **cloud-based monitoring** and optimizing models with **FPGA acceleration and quantization**.

ADDITIONAL EXPERIENCES

Lead Code Sensei **Code Ninjas** **02/2021 - 08/2021**

- Assisted ninjas aged 7-14 **develop games using block-based languages** (Scratch, MakeCode, etc), **object oriented languages** (Python, Java, JavaScript, etc), 3D modeling/printing with TinkerCAD, and game development with lua and unity.
- Worked with other sensei's to **develop various software's** to assist with the learning.

Researcher **Neuro Nexus** **09/2020 - Current**

- Alongside AltaML & Path Therapeutics, I worked towards: **developing and enhancing a ML model to be used in the detection of low, medium, and high level seizures.**
- Annotations for over 100 manually labelled EEG seizures were collected and utilized.
- A GUI was constructed for this product to be readily available within the neuroscience and mental health fields

Skills

Proficient: C++, python, Unix, Git, ROS, JavaScript, Algorithms & Data Structures, TensorFlow, React, Bash, API's, Embedded Systems, Docker, Git, System Verilog, PCB Design, CMake, Django, LLVM.

Expert: Java, Go, C#.NET, SQL Databases, OpenGL, Unity, Docker/Kubernetes, Cloud, Node.js, Flutter, Mips, x86.